# CSA1402 - COMPILER DESIGN

## Title: Compiler vs Interpreter

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# Introduction

## What is a Language Translator?

A language translator converts a high-level programming language into machine language so that the computer can understand and execute it.

There are three types of language translators:

- Compiler
- Interpreter
- Assembler

Example Languages

- Compiler → C, C++, Go
- Interpreter → Python, JavaScript, Ruby

# What is a Compiler?

**A compiler translates the entire source program into machine code before execution.**

## Features

- **Converts the whole program at once.**
- **Generates an executable (.exe) file.**
- **Detects errors after compilation.**
- **Execution is very fast.**
- **Suitable for large applications.**


## Examples

- **GCC (GNU Compiler Collection)**
- **Clang**
- **Microsoft Visual C++**

# What is an Interpreter?

**An interpreter translates and executes the program line by line.**

## Features

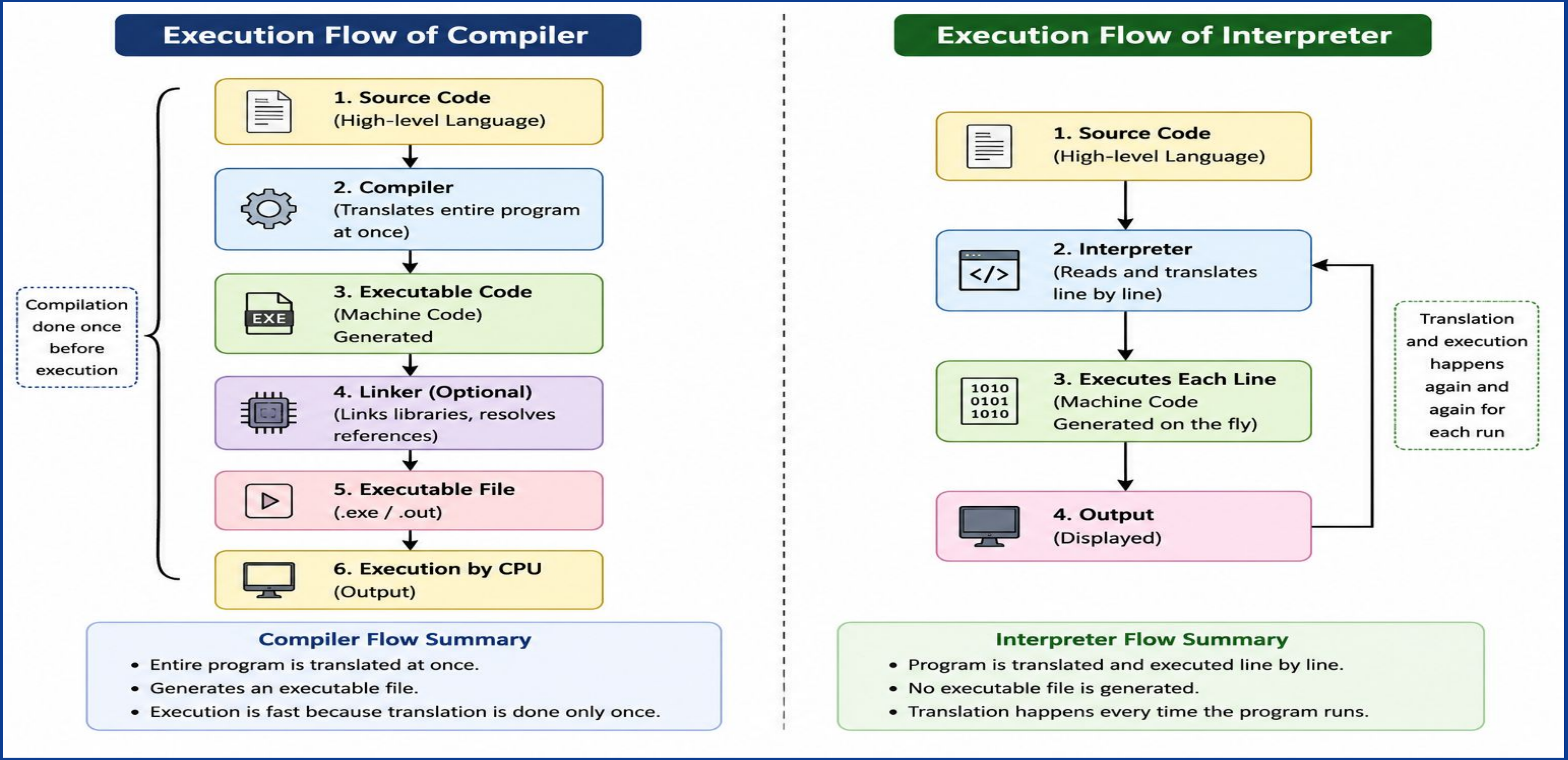
- No executable file is created.
- Stops immediately if an error occurs.
- Easier to debug.
- Slower execution.
- Good for scripting and testing.

## Examples

- Python Interpreter
- Node.js (JavaScript)
- Ruby Interpreter

# Compiler vs Interpreter

| Feature | Compiler | Interpreter |
|---|---|---|
| Translation | Translates entire program | Translates line by line |
| Execution Speed | Faster execution | Slower execution |
| Executable File | Generates executable file | No executable file |
| Error Reporting | Errors shown after compilation | Errors shown immediately |
| Used For | Used for C, C++ | Used for Python, JavaScript |


Execution Flow of Compiler
1. Source Code
(High-level Language)
2. Compiler
(Translates entire program at once)
3. Executable Code
EXE
(Machine Code) Generated
4. Linker (Optional)
(Links libraries, resolves references)
5. Executable File
(.exe / .out)
6. Execution by CPU
(Output)
Compilation done once before execution
Compiler Flow Summary
• Entire program is translated at once.
• Generates an executable file.
• Execution is fast because translation is done only once.
Execution Flow of Interpreter
1. Source Code
(High-level Language)
2. Interpreter
</>
(Reads and translates line by line)
3. Executes Each Line
1010
0101
1010
(Machine Code Generated on the fly)
4. Output
(Displayed)
Translation and execution happens again and again for each run
Interpreter Flow Summary
• Program is translated and executed line by line.
• No executable file is generated.
• Translation happens every time the program runs.

example code in C:

```
#include <stdio.h>
int main()
{
    int n;
    scanf("%d", &n);

    if (n % 2 == 0)
        printf("Even");
    else
        printf("Odd");

    return 0;
}
```

C Source Code (.c)

↓

Compiler (GCC)

↓

Executable File (.exe)

↓

Run Executable

↓

Output (Even/Odd)

Example code in python:

```python
# Program to check whether
 a number is odd or even

number = int(input("Enter an integer: "))

if number % 2 == 0:
    print(f"{number} is an Even number.")
else:
    print(f"{number} is an Odd number.")
```

Python Source Code (.py)

↓

Python Interpreter

↓

Reads One Line

↓

Executes That Line

↓

Reads Next Line

↓

Output (Even/Odd)

| COMPILER | INTERPRETER |
|---|---|
| **Advantages** | **Advantages** |
| 1. **Faster Execution** – Compiled programs run faster because the entire program is translated before execution. | 1. **Easy to Debug** – Errors are displayed immediately when encountered. |
| 2. **Creates Executable File** – Generates a standalone executable (.exe) that can be run without the compiler. | 2. **No Compilation Required** – Executes the program directly without creating an executable file. |
| 3. **Better Performance** – Optimizes the code for efficient use of CPU and memory. | 3. **Platform Independent** – The same source code can run on different platforms if the appropriate interpreter is installed. |
| 4. **Suitable for Large Applications** – Best for developing operating systems, games, and desktop software. | 4. **Faster Development** – Suitable for testing, scripting, and rapid application development. |
| **Disadvantages** | **Disadvantages** |
| 1. **Compilation Takes Time** – The entire program must be compiled before it can run. | 1. **Slower Execution** – The program is translated and executed line by line. |
| 2. **Difficult to Debug** – Errors are reported after compilation, making debugging less convenient. | 2. **No Executable File** – An interpreter is required every time the program runs. |
| 3. **Platform Dependent** – Executable files are usually created for a specific operating system or processor. | 3. **Less Efficient** – Translation happens each time the program is executed, reducing performance. |
| 4. **Requires More Memory** – Compilation and executable files consume more storage and memory. | 4. **Not Suitable for Large Applications** – Slower execution makes it less ideal for performance-critical software. |

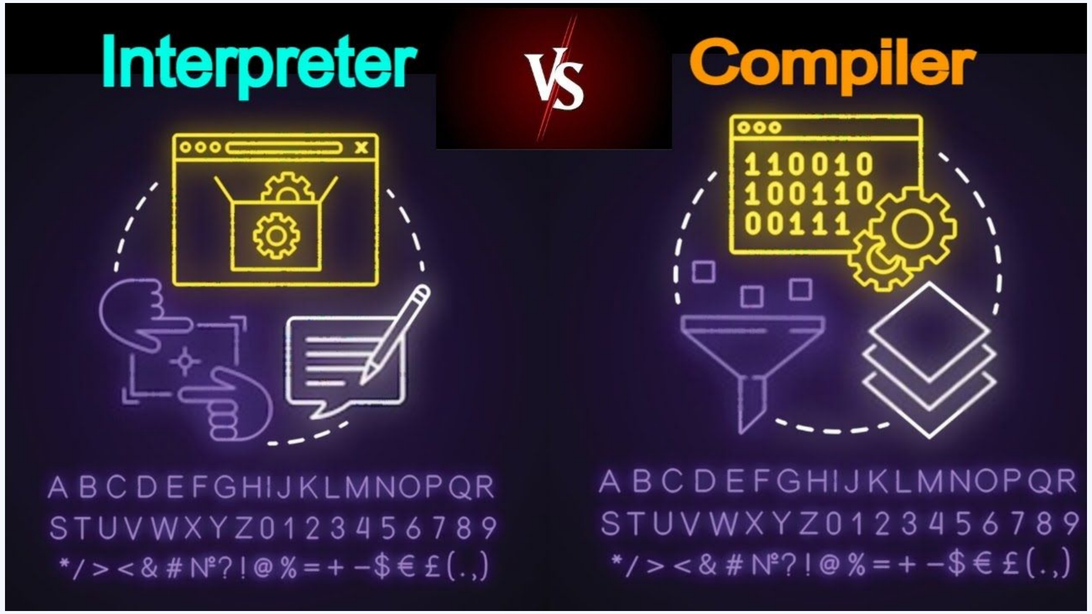
# Where Are They Used?

## Compiler

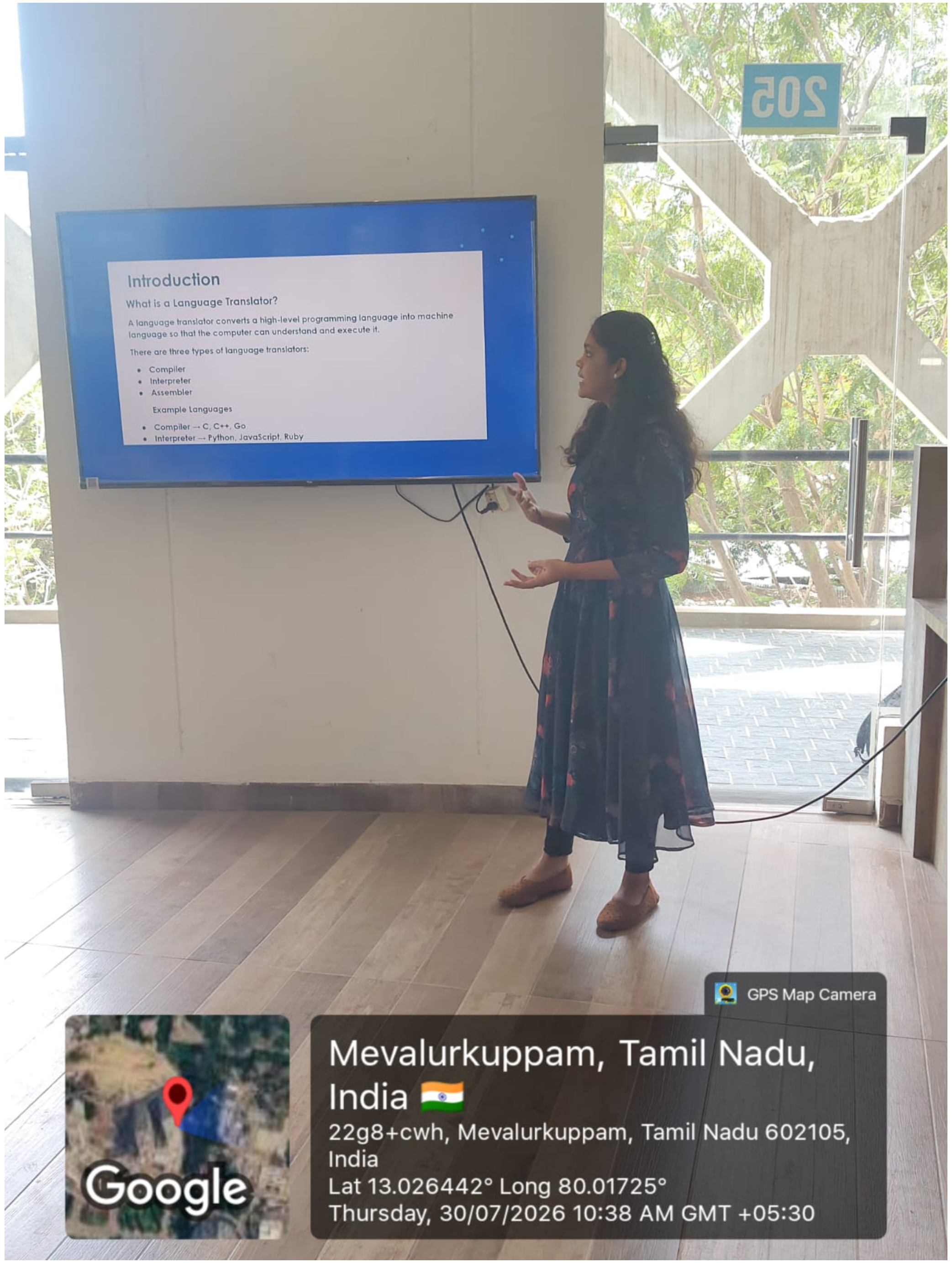
- Operating Systems
- Games
- Desktop Applications
- Embedded Systems

## Interpreter

- Web Development
- Data Science
- Artificial Intelligence
- Automation Scripts



# THANK YOU!



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